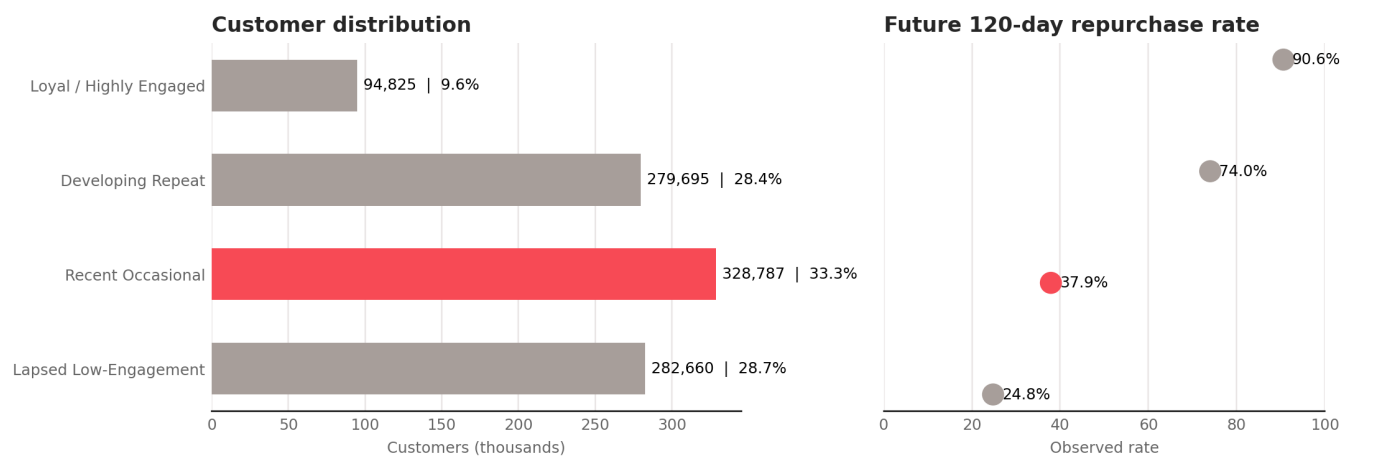


H&M Customer Retention & Predictive CRM Analytics

Growing repeat purchase through customer segmentation, propensity modeling, and targeted CRM prioritization

PORTFOLIO CASE STUDY BY CHLOE LIU | SEPTEMBER 2026

31M+	985,967	4	120 days
Transactions	Customers analyzed	Lifecycle segments	Prediction horizon



Data source: H&M Personalized Fashion Recommendations dataset. Transaction price is a normalized spend index, not a currency amount.

01 Executive summary

From broad segmentation to an operational weekly CRM scoring workflow

Business question

How can H&M identify Recent Occasional customers who are likely—or unlikely—to purchase again in the next 120 days, so CRM resources can be allocated more intelligently?

328,787

Recent Occasional customers

37.9%

Observed repurchase rate

65.3%

Top-decile rate

16.4%

Bottom-decile rate

What the analysis found

- RFM-based K-Means revealed four interpretable lifecycle segments. Recent Occasional was the largest segment and sat between Developing Repeat and Lapsed Low-Engagement in both engagement and future purchase behavior.
- Behavioral signals—membership, Fashion News engagement, product exploration, and channel usage—improved rank ordering beyond an RFM-only baseline.
- The selected Logistic Regression achieved 0.685 ROC AUC and 0.556 PR AUC on an untouched test set. A nonlinear challenger improved only slightly, supporting Logistic Regression as the operational primary model.
- The selected model separated the top and bottom propensity deciles by 48.9 percentage points, creating a practical basis for differentiated CRM treatment.

Recommended operating model

Score eligible Recent Occasional customers on a recurring cadence, assign them to propensity tiers, apply different message intensity and offer depth, and validate incremental impact through randomized holdout tests.

02 Data and analytical design

A time-ordered design prevents future behavior from leaking into historical features

Dataset

- 31M+ transaction rows linked to customer and article attributes.
- 985,967 customers with at least one purchase in the 364-day historical observation window.
- The available price field is normalized; it is interpreted as a spend index.

Observation and outcome windows

Period	Purpose	Data used
364-day historical window	Build RFM segments and customer features	Purchases, product mix, channel use, membership and Fashion News indicators
Subsequent 120-day window	Define the target	1 if the customer purchased again; 0 otherwise

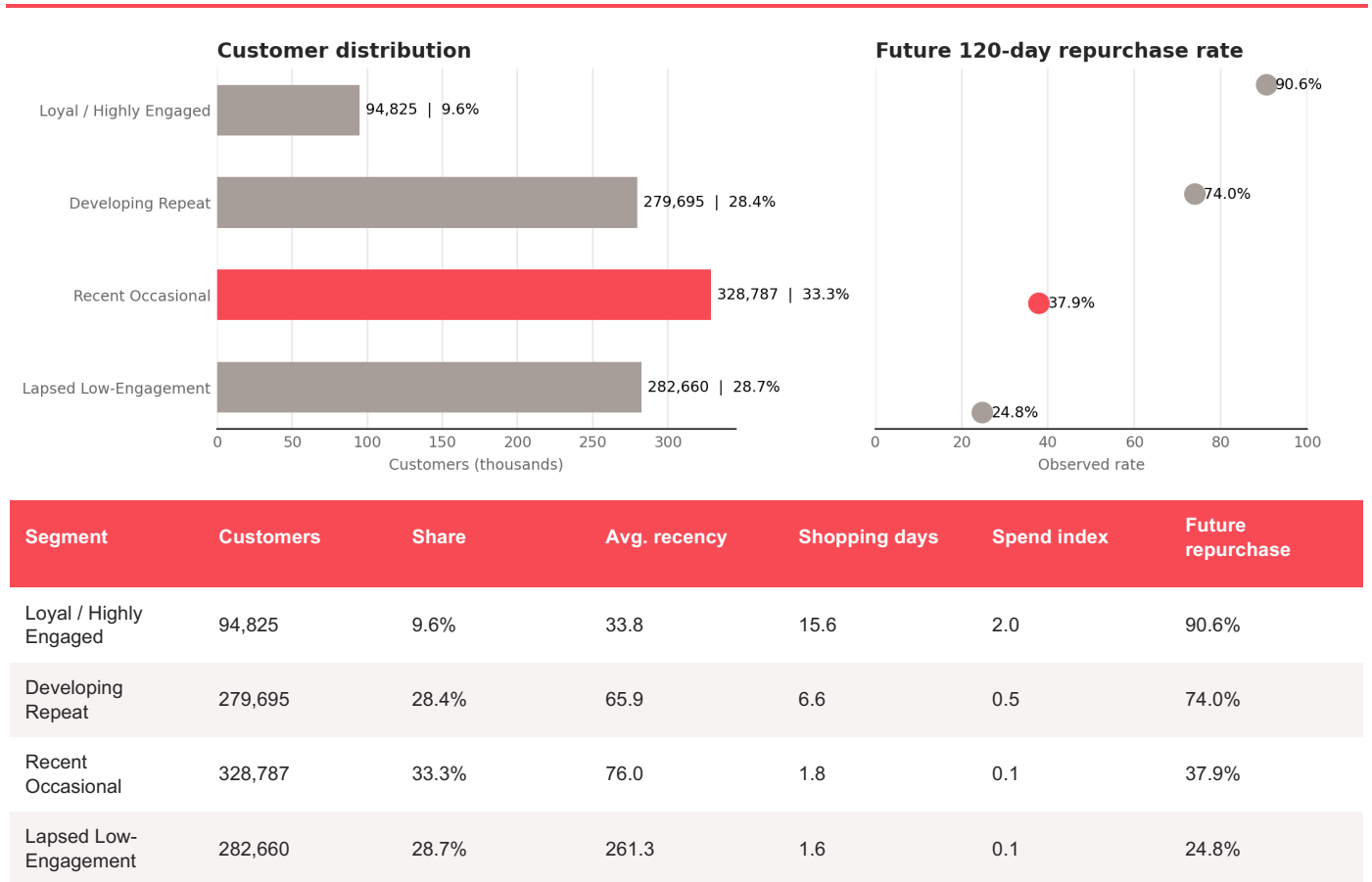
End-to-end workflow

Step	Question	Method
1. Segment	Who are the customers?	RFM feature engineering + K-Means
2. Prioritize	Which segment is the growth opportunity?	Segment size + future repurchase comparison
3. Profile	What signals distinguish stronger repeat behavior?	Descriptive profiling, SMD, redundancy review
4. Predict	Who is likely to repurchase within 120 days?	Logistic Regression + nonlinear challenger
5. Activate	How should CRM treatment differ?	Propensity deciles + test-and-learn strategy

The future-purchase label was attached only after clustering. It was not used to create the customer segments.

03 Customer segmentation

Four lifecycle groups create a clear retention landscape



Why focus on Recent Occasional?

It is the largest segment, remains relatively recent, but has not established repeat-shopping depth. Its 37.9% future repurchase rate confirms a mixed-outcome population: some customers will return naturally, while others are moving toward lapse. That ambiguity makes the segment both commercially important and suitable for propensity scoring.

04 Customer profiling and feature selection

Behavioral signals were screened before modeling

Recent Occasional customers were compared with Developing Repeat customers to identify interpretable signals associated with stronger repeat behavior. The goal was to build a compact, non-redundant feature set—not to include every available variable.

Feature families reviewed

Feature family	Examples reviewed	Decision
RFM controls	Recency, shopping days, spend index	Retained as recency, log frequency, and log monetary
Customer engagement	Active status, club status, Fashion News indicators	Retained active club member and regular Fashion News
Purchase intensity	Item count, items per shopping day, distinct articles	Basket depth tested incrementally
Product exploration	Distinct garment/product/index groups, entropy	Retained product-group entropy
Channel behavior	Channel-only flags, both channels, Channel 2 share	Retained both-channel usage; Channel 2 share tested incrementally

Selection logic

- Standardized mean differences (SMDs) identified features with meaningful profile separation.
- Within-family correlation checks removed redundant representations, such as overlapping engagement flags and closely related product-diversity measures.
- Shopping days and spend index were not duplicated as candidate behavioral signals because their transformed versions already served as RFM controls.
- Final candidates remained grounded in CRM interpretability and availability at scoring time.

Profile differences and model coefficients are associations and predictive signals; they do not establish that changing a feature will cause repurchase.

05 Predictive model design

Validation selects the specification; the untouched test set estimates final performance

Modeling population and split

328,787	197,271	65,758	65,758
Recent Occasional customers	Training	Validation	Test

The target is whether a Recent Occasional customer purchased in the subsequent 120-day outcome window. The positive rate remained approximately 37.9% across all three partitions.

Candidate Logistic Regression specifications

Model	Included features
RFM baseline	Recency + log frequency + log monetary
Expanded behavioral	RFM + regular Fashion News + active club membership + product-group entropy + both-channel usage
Expanded + basket depth	Expanded behavioral + items per shopping day
Expanded + channel preference	Expanded behavioral + Channel 2 item share
Expanded + both candidates	Expanded behavioral + both incremental features

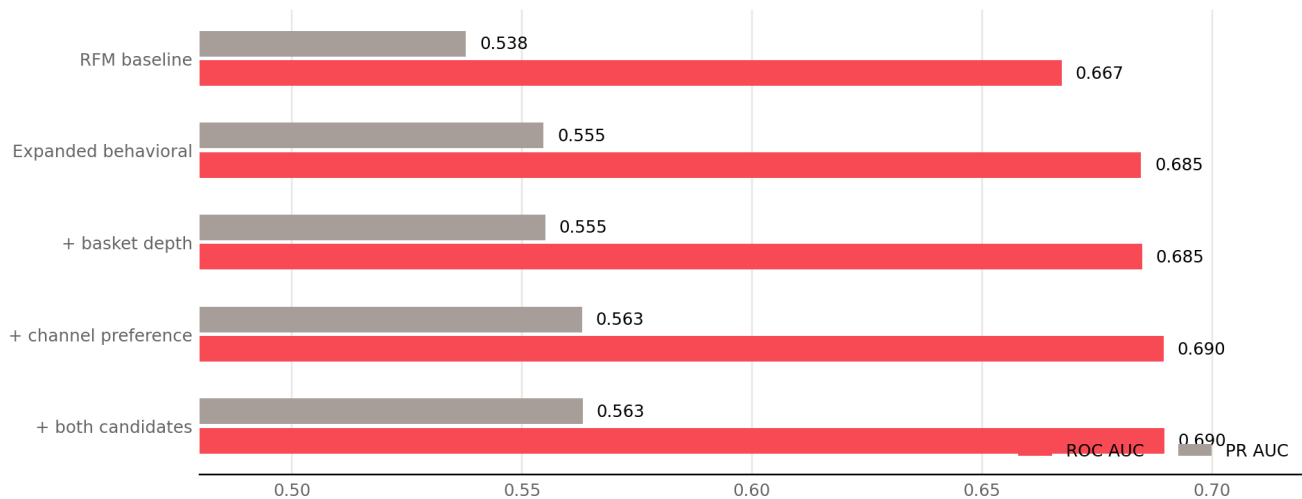
Evaluation metrics

- ROC AUC evaluates ranking quality across every possible classification threshold.
- PR AUC emphasizes identification of actual purchasers and is useful when class balance matters.
- Brier score evaluates probability accuracy; lower values indicate better-calibrated predictions.

A fixed yes/no threshold is not required to calculate AUC. For CRM, ranked probabilities and budget-based deciles are more useful than a universal 50% cutoff.

06 Logistic model selection

Channel preference adds the clearest incremental value



Selection decision

The Expanded + Channel Preference model was selected. It improved validation ROC AUC from 0.667 to 0.690 and PR AUC from 0.538 to 0.563. Adding basket depth on top produced only a negligible gain, so the more parsimonious specification was retained.

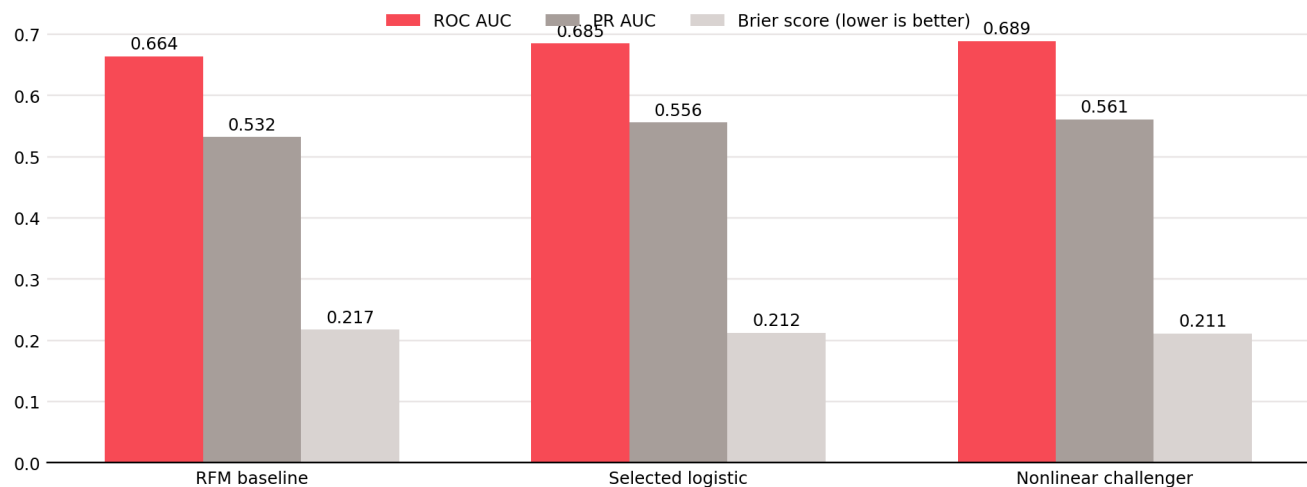
Estimated Logistic Regression equation

$$\begin{aligned} \text{logit}(p) = & -1.378 - 0.233 \, z(\text{Recency}) + 0.406 \, z(\text{Log Frequency}) + 0.025 \, z(\text{Log Monetary}) \\ & + 0.326 \, \text{Regular Fashion News} + 0.736 \, \text{Active Club Member} + 0.121 \, z(\text{Product-Group Entropy}) \\ & + 0.234 \, \text{Both Channels} - 0.181 \, z(\text{Channel 2 Item Share}) \end{aligned}$$

Here, p is the probability that a Recent Occasional customer purchases again during the future 120-day window. Continuous variables were standardized before model fitting.

07 Final test performance

The selected logistic model improves ranking and probability quality on unseen customers



Model	ROC AUC	PR AUC	Brier score	Role
RFM baseline	0.6640	0.5319	0.2173	Reference
Selected logistic	0.6851	0.5559	0.2122	Operational primary
Nonlinear challenger	0.6887	0.5608	0.2113	Challenger / diagnostic

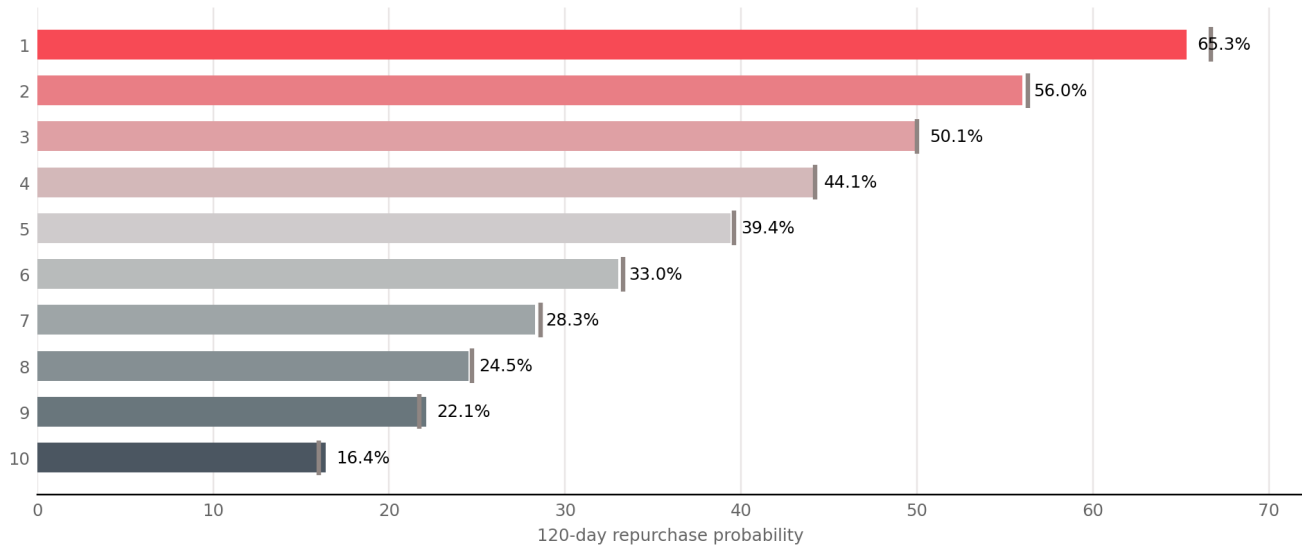
Interpretation

- Behavioral and channel signals add predictive information beyond RFM alone.
- The nonlinear challenger delivers only a modest lift over Logistic Regression.
- Logistic Regression remains the recommended production model because it is easier to explain, audit, score, and connect to CRM decision rules.

The nonlinear model is valuable as a challenger and for identifying nonlinear patterns; it does not materially change the operational decision.

08 Decile validation

The model creates a strong, monotonic prioritization ladder



65.3%

Top-decile observed rate

16.4%

Bottom-decile observed rate

48.9 pp

Top-to-bottom spread

1.72×

Top-decile lift

Each decile contains approximately 6,576 test customers. Decile 1 represents the highest predicted propensity and Decile 10 the lowest. Bars show observed repurchase rates; vertical markers show average predicted probabilities. The close alignment supports using scores as probabilities as well as rankings.

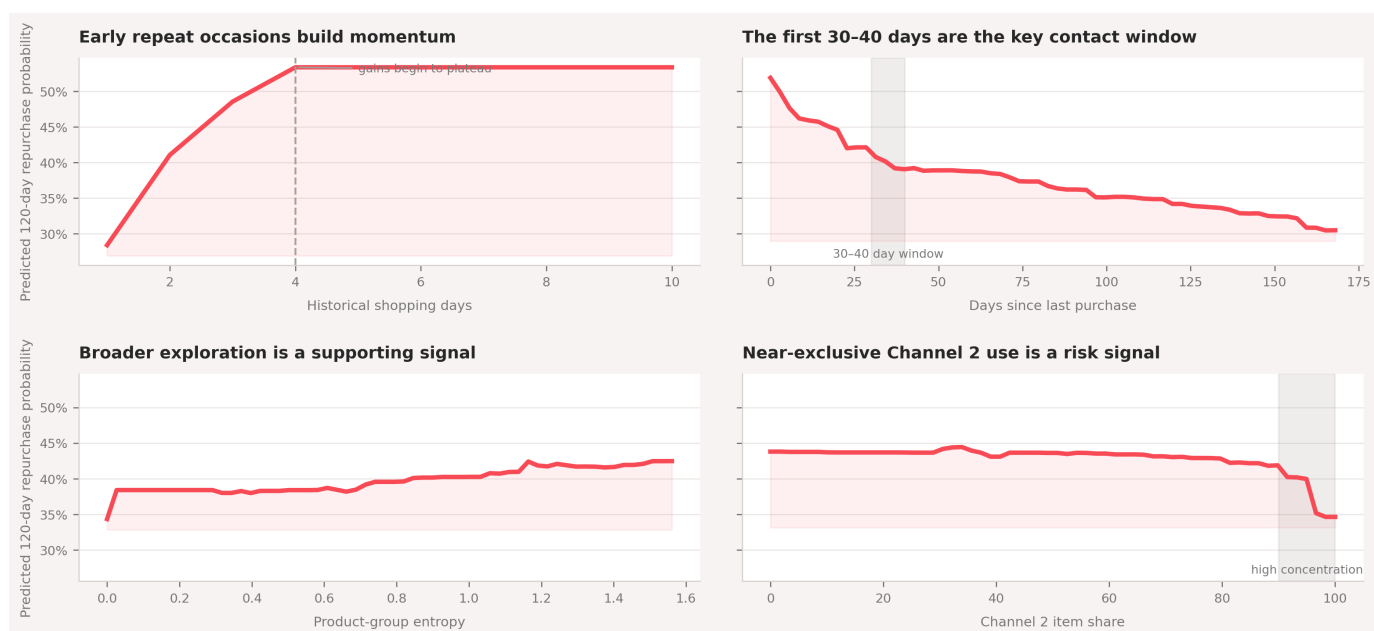
Business meaning

The model does not need to declare every customer a purchaser or non-purchaser. It creates a ranked audience that can be matched to available budget, contact policy, and offer economics.

09 What the nonlinear challenger reveals

Shopping frequency, recency, and channel concentration carry the strongest behavioral patterns

Signal	Observed nonlinear pattern	CRM implication
Historical shopping days	Predicted repurchase rises sharply through roughly four shopping days, then incremental gains begin to plateau; customers above four days are uncommon in this cohort.	Early repeat occasions are valuable milestones for building shopping habit.
Recency	The steepest deterioration occurs during roughly the first 30–40 days since last purchase, followed by a more gradual decline.	Use recency-sensitive contact windows rather than waiting for deep lapse.
Product-group entropy	Broader product exploration is positively related to repurchase, with a smaller effect than frequency and recency.	Use cross-category discovery as a supporting—not primary—activation lever.
Channel 2 item share	Mixed channel shares are relatively stable, while predicted repurchase falls sharply near complete Channel 2 concentration.	Treat Channel 2-only behavior as a risk signal and test cross-channel onboarding.



Observed channel outcomes

57.3%

Both channels

44.4%

Channel 1 only

31.6%

Channel 2 only

These patterns are partial-dependence and descriptive findings, not intervention effects. Channel identities are not labeled in the public dataset, so recommendations use Channel 1/2 terminology.

10 Predictive CRM strategy

Use the score to vary effort, message, and incentive—not to exclude customers permanently

Audience	Observed signal	Recommended treatment	Primary test metric
High propensity (Deciles 1–3)	50.1%–65.3% observed repurchase	Low-cost nurture: new arrivals, personalized recommendations, loyalty progress; suppress unnecessary discounts	Incremental revenue per contacted customer
Mid propensity (Deciles 4–7)	28.3%–44.1% observed repurchase	Habit-building: category discovery, cross-channel education, timed reminders, light incentive tests	Incremental 120-day repurchase rate
Low propensity (Deciles 8–10)	16.4%–24.5% observed repurchase	Selective win-back: fewer but stronger messages, value-led offer, preference refresh; maintain a no-contact holdout	Incremental margin and unsubscribe rate

Testing framework

- Randomize treatment and holdout within each propensity tier so uplift is measured against the correct baseline.
- Track incremental repurchase, incremental margin, revenue per message, unsubscribe rate, and discount cost—not raw conversion alone.
- Refresh scores on a recurring cadence and monitor decile-level calibration, audience volume, and performance drift.
- Use contact-policy guardrails so low propensity does not automatically translate into message volume or aggressive discounting.

The model answers who to prioritize and how much natural return tendency to expect. Experiments determine which CRM action actually causes incremental repurchase.

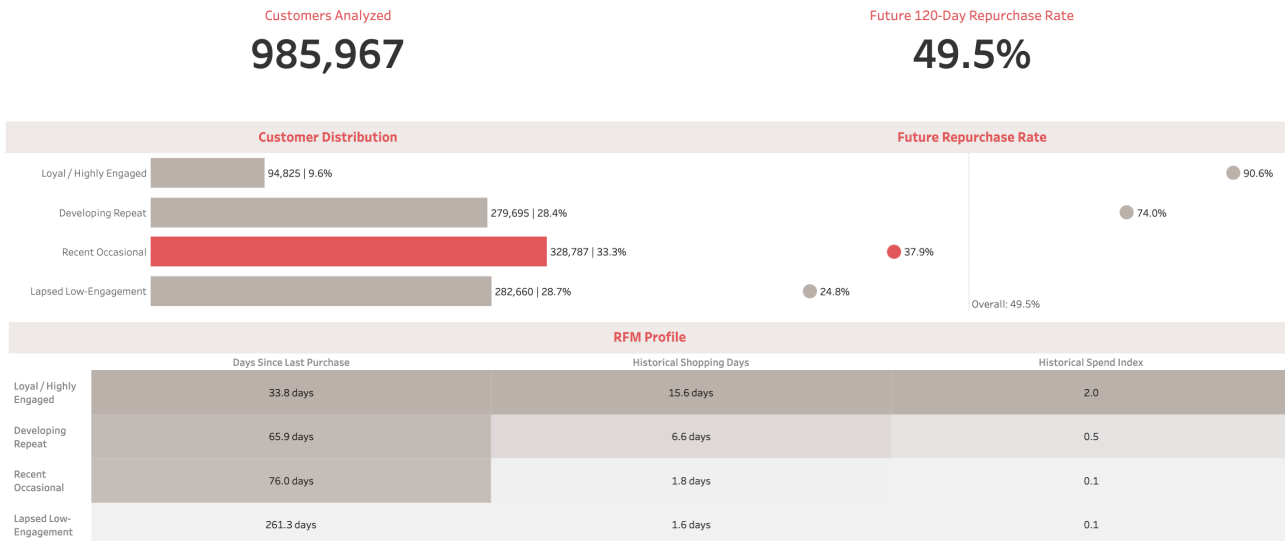
11 Interactive dashboards

Two Tableau views connect customer strategy with operational prioritization

Dashboard 1 — Customer Segments & Retention Overview

H&M Customer Segments & Retention Overview

Historical customer behaviour and subsequent 120-day repurchase performance



Dashboard 2 — Repurchase Propensity & Predictive CRM

H&M Repurchase Propensity & Predictive CRM Dashboard

Identifying high- and low-propensity Recent Occasional customers for targeted retention strategies

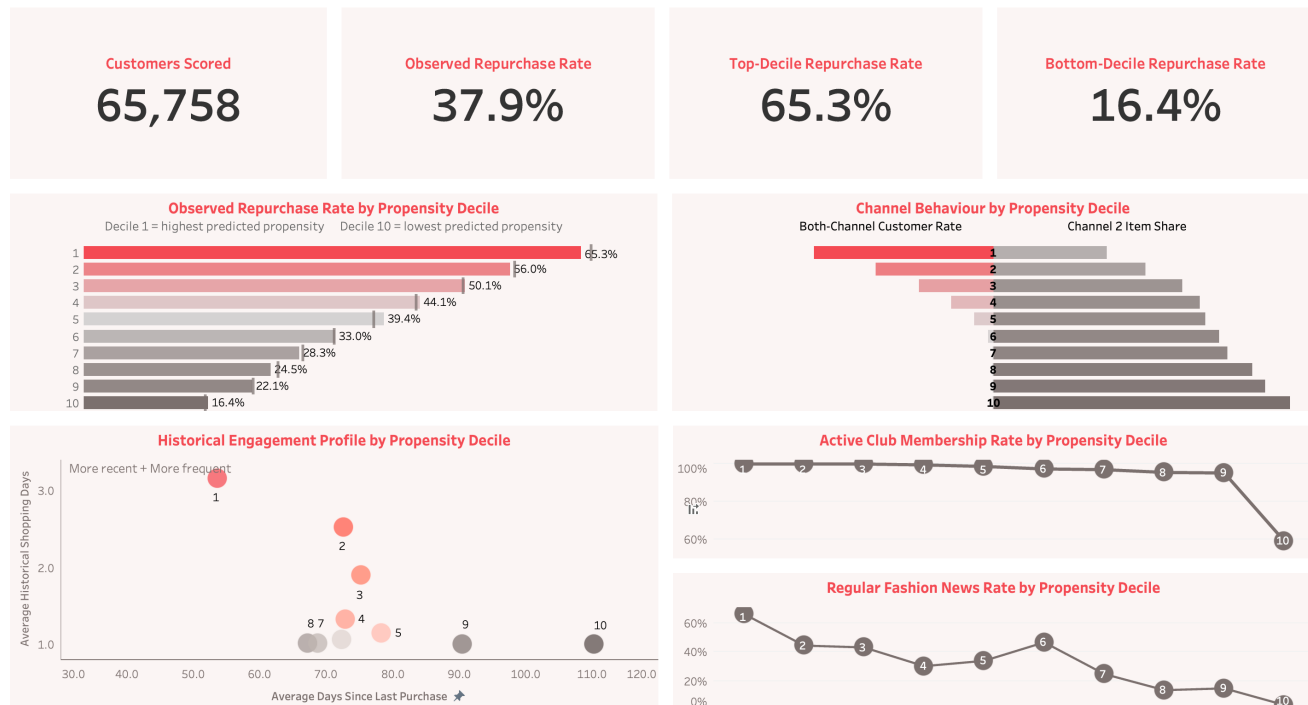


Tableau Public:

https://public.tableau.com/app/profile/chloe.liu3692/viz/HMDashboard_17890951816630/HMRepurchasePropensityPredictiveCRMDashboard

12 Implementation notes and project assets

A reproducible portfolio workflow across SQL, Python, machine learning, and Tableau

Technical workflow

Layer	Purpose
SQL	Create the 364-day historical customer table, RFM features, outcome label, entropy, and customer-level profiling features
Python — K-Means	Scale RFM features, evaluate K with the elbow method, fit four segments, and attach future outcomes after clustering
Python — Profiling	Compare Developing Repeat and Recent Occasional, calculate SMDs, review redundancy, and create model inputs
Python — Logistic Regression	Use train/validation/test splits, compare specifications, estimate the final interpretable model, and export scored customers
Python — Nonlinear model	Fit a HistGradientBoosting challenger and inspect partial-dependence patterns
Tableau	Deliver segment-level and propensity-level interactive dashboards for business users

Final business takeaway

H&M does not need to treat every Recent Occasional customer the same. A compact, interpretable propensity model can identify meaningfully different natural repurchase likelihoods, protect incentive budget, and create a disciplined foundation for tier-specific CRM experimentation.